Cauchy-Schwarz Inequality

Statement: For any two random variables X and Y, we have

$$|E(XY)| \le \sqrt{E(X^2)E(Y^2)},$$

with equality if and only if $P(X=\alpha Y)$ for some constant $\alpha \in \mathbb{R}$.

Proof: Consider the random variable $W=(X-\alpha Y)^2$. Clearly, W is a non-negative random variable for any $\alpha \in \mathbb{R}$. Thus we have,

$$0 \le E(W) = E[(X-\alpha Y)^2]$$

$$= E[X^2 - 2\alpha XY + \alpha^2 Y^2]$$

$$= E(X^2) - 2\alpha E(XY) + \alpha^2 E(Y^2).$$

Take $\alpha = \dfrac{E(XY)}{E(Y^2)}$. Then, we have,

$$0 \le E(X^2) - 2\frac{(E(XY))^2}{E(Y^2)} + \frac{(E(XY))^2}{E(Y^2)}$$

$$\Rightarrow E(X^2) - \frac{(E(XY))^2}{E(Y^2)} \ge 0$$

$$\Rightarrow (E(XY))^2 \le E(X^2)E(Y^2)$$

$$\Rightarrow |E(XY)| \le \sqrt{E(X^2)E(Y^2)}.$$

Now if $\quad |E(XY)| = \sqrt{E(X^2)E(Y^2)}$

$$\Rightarrow \quad (E(XY))^2 = E(X^2)E(Y^2)$$

$$\Rightarrow \quad E(X^2) - \frac{(E(XY))^2}{E(Y^2)} = 0$$

$$\Rightarrow \quad E(X^2) - 2\frac{E(XY)}{E(Y^2)} \cdot E(XY) + \frac{(E(XY))^2}{(E(Y^2))^2} E(Y^2) = 0$$

$$\Rightarrow \quad E\left(X - \frac{E(XY)}{E(Y^2)} Y\right)^2 = 0$$

$$\Rightarrow \quad X = \frac{E(XY)}{E(Y^2)} Y \quad \text{with probability } 1.$$